EPINEON AI

PRECISION LIFE SCIENCE COMPANY

TECHNICAL CHALLENGE

Internship PFE 2026 · ESITH Cohort

LLM Monitoring System

Duration	1 week (5 working days) from receipt of this document
Recipients	Pre-selected candidates — ESITH Cohort 2025/2026
Deliverables	GitHub repository · PDF Report · Functional Demo
Contact	Email: recrutement@epineon.ai Subject: EPINEON Challenge | LLM Monitoring System | [Your Name]
Position reference	Backend & AI/ML · Data Engineering · Full-Stack

01. Context & Objective

Epineon AI develops high-impact artificial intelligence solutions in the Life Science domain. As part of our internal product roadmap, we continuously monitor the evolution of the large language model (LLM) ecosystem to always recommend the best option to our enterprise clients.

Dozens of new models are released every week. Choosing the right model for a given use case — coding, reasoning, autonomous agents, RAG — has become a strategic challenge. This is the LLM Monitoring System: an automated system for collecting, normalizing, scoring, and recommending LLMs.

Your mission

Build a simplified but fully functional version of the LLM Monitoring System, comprising:

- A data collection pipeline from public LLM benchmark sources
- A composite scoring and recommendation engine by enterprise profile
- A visualization and reporting module
- (Bonus) An automation or daily monitoring agent pipeline

System architecture

The three modules are sequential and feed into one another:

MODULE 1
Collection
Scraping · API · Storage

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MODULE 2
Scoring
Formula · Profiles · API

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MODULE 3
Visualization
Dashboard · Digest Report

02. The 3 Modules

Module 1 **Data Collection & Normalization**	
T1	Select and query at least 2 sources from: Artificial Analysis (artificialanalysis.ai), HuggingFace Open LLM Leaderboard, LLM Stats (llm-stats.com), Vellum AI Leaderboard or any similar platforms.
T2	Extract for each model: name, Intelligence score, price/token (input & output), speed (tokens/s), TTFT latency, context window size, and license type (Apache / MIT / Proprietary).
T3	Normalize all metrics to a consistent scale and store the data in a database of your choice (PostgreSQL, SQLite, Supabase, or MongoDB).
T4	Implement a new model detection function that identifies models absent from the previous collection run.
T5	Document the sources used, their update frequency, and any limitations encountered.

Module 2 **Scoring & Recommendation Engine**	
T1	Design and implement a composite scoring model for each collected LLM. You are free to define your own formula and weighting logic — justify your choices based on what matters most for enterprise use cases (reasoning, performance, cost, speed, latency, etc.).
T2	Implement an enterprise profile system with at least 4 profiles, for example: Coding/Dev, Reasoning/Analysis, Enterprise Agents, Long Context RAG, Multilingual, or Minimum Cost.
T3	For each profile, adapt your scoring weights accordingly and return the top 3 recommended models with a clear justification.
T4	Add a compliance filter: exclude models with non-commercial licenses when the 'commercial use' option is enabled.
T5	Expose this logic via a REST API with an endpoint such as /recommend?profile=coding&commercial=true.

Module 3 **Visualization & Digest Report**	
T1	Build a dashboard or visualization interface allowing users to: compare scores across multiple models, filter by enterprise profile, and view newly detected models.
T2	Auto-generate a synthetic digest report (Markdown or PDF) containing: Top 5 models by profile, significant movements (new models, price drops), and recommendations with justification.
T3	Technology choice is free: React.js, Power BI, Streamlit, Plotly Dash, or any other tool.
T4	The report must be generated automatically from your database without manual intervention.

Bonus Module — Automated Monitoring Agent

Automate the full pipeline — collection, scoring, and reporting — using a scheduler or agent:

- Python scheduler (APScheduler or Celery) with daily execution
- N8N workflow with data collection, transformation, and email/Slack delivery
- LangGraph or CrewAI agent orchestrating the collection and analysis steps
- Bash script + cron job with automatic digest report generation

The bonus is appreciated but not required. It can significantly differentiate your application.

03. Expected Deliverables

Send your submission to recrutement@epineon.ai — Subject: EPINEON Challenge| LLM Monitoring System | [Your Name]

#	Deliverable	Expected content
L1	**GitHub Repository**	Complete source code with a detailed README (setup, usage, architecture). Repository must be public or shared with the recruiter.
L2	**Technical Report (PDF, 3–5 pp)**	Justified technical choices, system architecture, challenges encountered, limitations and improvements. Include screenshots.
L3	**Functional Demo**	Deployed URL of your dashboard/API, or a screen recording (3–5 min). The demo must showcase all 3 modules in action.
L4	**Oral Presentation (if selected)**	10-minute presentation + 5 minutes of Q&A before the Epineon team. Be ready to explain every technical decision.

04. Evaluation

Your work will be assessed across 5 dimensions. Total score: 100 points.

Criterion	Pts	What we look for
Code quality & architecture	25	Readability, structure, modularity, error handling, documentation. Git commit history, .env for credentials, adherence to best practices.
Functional completeness	25	All 3 modules implemented and working. Results consistent with source data. API responds correctly to all required queries.
Business analysis & relevance	10	Recommendations are justified. Shows understanding of enterprise LLM trade-offs: cost, licensing, latency, GDPR compliance.
Presentation & communication	20	Clarity of technical report, demo quality, ability to explain choices simply. Logical and structured thinking.
Creativity & initiative	20	Bonus module implemented, additional relevant features, original approach, or unexpected insight from the LLM data.

Qualification threshold: 80 / 100

Candidates reaching this threshold will be invited to a 15-minute oral presentation for Epineon AI team. In case of a tie, the bonus module and code quality are the deciding criteria.

05. Constraints & Rules

Technology stack

You are free to choose your tools, subject to the following constraints:

- Primary language: Python (mandatory for backend logic and scoring)
- Database: PostgreSQL, SQLite, MongoDB or Supabase — document your choice and reasoning
- API: FastAPI or Flask recommended — any other REST framework accepted
- Visualization: React.js, Streamlit, Plotly Dash, Power BI, or any tool of your choice
- Version control: Git is mandatory — a meaningful commit history is expected, not a single final commit

Authorized data sources

Use at least 2 of the following either using API if available or Scrapping them :

- Artificial Analysis — artificialanalysis.ai/leaderboards/models
- HuggingFace Open LLM Leaderboard — free public API
- LLM Stats — llm-stats.com
- Vellum AI Leaderboard — vellum.ai/llm-leaderboard
- Onyx LLM Leaderboard — onyx.app/llm-leaderboard

Minimum requirement: data for 20 distinct LLM models with at least 8 metrics each.

Integrity rules

- This challenge is individual. Each candidate must submit their own work.
- AI assistants (ChatGPT, Claude, Copilot, etc.) are allowed and encouraged — but you must be able to explain and defend every line of your code during the oral presentation.
- Cite your sources: if you reuse open-source code, mention it in your README.
- Fabricated or invented data is not accepted — use real data from the listed sources.

06. Timeline & Submission

Day	Milestone	Expected output
D0	**Receipt**	Read this document carefully and send a confirmation email to recrutement@epineon.ai
D1–D2	**Setup & Collection**	Git repository created, environment configured, Module 1 implemented (data collection + storage)
D3	**Scoring & API**	Module 2 implemented — composite scoring formula and recommendation API
D4	**Dashboard & Report**	Module 3 implemented — visualization dashboard and auto-generated digest report
D4	**Bonus (optional)**	Automation module implemented if time allows
D5	**Final submission**	Submit by email before 11:59 PM — GitHub link, PDF report, and demo URL

How to submit

Send your submission to recrutement@epineon.ai with the exact subject line:

> **EPINEON Challenge | LLM Monitoring System | [First Name LAST NAME]**

Your email must include:

- Link to your GitHub repository (public or shared with the recruiter)
- Technical report attached as a PDF (3–5 pages)
- URL of your deployed dashboard or API
- A short paragraph (3–5 lines) describing what you found most interesting or challenging

Good luck!

The Epineon AI Team